Eric A. Dunipace

Psychiatry · Biostatistics · Causal Inference · Optimal Transport

edunipace@post.harvard.edu
ericdunipace
https://ericdunipace.github.io

Education

2023 **Doctor of Medicine**, *David Geffen School of Medicine*, UCLA

2021 **Doctor of Philosophy**, *Biostatistics*, Harvard University
Thesis: *Optimal Transport Methods for Causal Inference, Multisample Testing, and Model Interpretation*. Advisors: José R. Zubizarreta and Lorenzo Trippa.

2019 **Master of Arts**, *Biostatistics*, Harvard University

2014 **Master of Science**, *Global Health and Population*, Harvard University
Thesis: *Estimating the Causal Effects of the Maternal Health Voucher Program in Bangladesh*. Advisor: Goodarz Danaei.

2008 **Bachelor of Arts with Honors**, *Molecular and Cell Biology*, University of California, Berkeley
Thesis: *Role of Naip5 and Flagellin in Innate Immune Responses to* Legionella pneumophila *and* Salmonella typhimurium. Advisor: Russell Vance.

Postgraduate Training

2025–Present **Resident Physician**, *Department of Psychiatry and Biobehavioral Sciences*, UCLA

2023–2025 **Resident Physician**, *Department of Psychiatry and Behavioral Sciences*, Stanford University

Publications

2025 **Dunipace**, **E. A.**, & Trippa, L. (2025). Interpretable model summaries using the Wasserstein distance. *The American Statistician*, *79*(4), 421–434. https://doi.org/10.1080/00031305.2025.2551223

2022 Sumayin Ngamdu, K., Mallawaarachchi, I., **Dunipace**, **E. A.**, Chuang, L.-H., Jafri, S. H., Shah, N. R., Jeong, Y. N., Morrison, A. R., & Bhatt, D. L. (2022). Association between periodontal disease and cardiovascular disease (from the NHANES). *American Journal of Cardiology*, *178*, 163–168. https://doi.org/10.1016/j.amjcard.2022.05.028

2018 Huang, W. A., **Dunipace**, **E. A.**, Sorg, J. M., & Vaseghi, M. (2018). Liver Disease as a Predictor of New Onset Atrial Fibrillation. *Journal of the American Heart Association*, *7*(15). https://doi.org/10.1161/JAHA.118.008703

2014 Moussally, J., **Dunipace**, **E. A.**, & Fu, R. (2014). TraumaLink: Helping People Survive Traffic Accidents in Bangladesh. In *Health and South Asia* (pp. 67–70). The Lakshmi Mittal South Asia Institute at Harvard University.

2011 Lightfield, K., Persson, J., Trinidad, N., Brubaker, S., Kofoed, E., Sauer, J.-D., **Dunipace**, **E.**, Warren, S., Miao, E., & Vance, R. (2011). Differential requirements for NAIP5 in activation of the NLRC4 inflammasome. *Infection and Immunity*, *79*(4). https://doi.org/10.1128/IAI.01187-10

2008 Lightfield, K., Persson, J., Brubaker, S., Witte, C., von Moltke, J., **Dunipace**, **E.**, Henry, T., Sun, Y.-H., Cado, D., Dietrich, W., Monack, D., Tsolis, R., & Vance, R. (2008). Critical function for Naip5 in inflammasome activation by a conserved carboxy-terminal domain of flagellin. *Nature Immunology*, *9*(10). https://doi.org/10.1038/ni.1646

Preprints

2022 **Dunipace, E. A.** (2022). Optimal transport weights for causal inference. *ArXiv210901991 Cs Econ Stat*. Retrieved September 7, 2021, from http://arxiv.org/abs/2109.01991

Software

R packages
- `WpProj`: Wasserstein projections for interpretable models
- `causalOT`: optimal transport methods for weighting in causal inference
- `approxOT`: approximate optimal transport library
- `graphTest`: graph-based multisample testing
- `RcppCGAL`: header files for the CGAL library

Invited Presentations

Feb. 2023 *Optimal Transport Weights for Causal Inference*, Missing Values & Causal Inference Virtual Seminar. École Polytechnique and Université de Montpellier. Paris/Montpellier, France.

Dec. 2022 *Optimal Transport Weights for Causal Inference*, CMStatistics 2022. King's College, London, UK

Oct. 2021 *Interpretable Posterior Summaries Using the Wasserstein Distance*, Research Frontiers in Biomathematics seminar series. UCLA. Los Angeles, CA

Conference Presentations

2022 *Cavernous Sinus Metastasis of p16-Tonsillar Squamous Cell Carcinoma: A Diagnostic Challenge*, Combined Section Meetings of the Triological Society. Coronado, CA

2022 *Optimal Transport Weights for Causal Inference*, Joint Statistical Meetings. Washington, DC

2020 *Optimal Transport Weights for Causal Inference*, Joint Statistical Meetings. Online

2019 *Interpretable Posterior Summaries Using the Wasserstein Distance*, Joint Statistical Meetings. Denver, CO

2015 *Connecting Californians to Care*, Network of Ethnic Physician Organization Summit. Riverside, CA

Awards and Scholarships

2018 Certificate of Distinction in Teaching, Department of Biostatistics, Harvard University

2014 David Geffen Scholarship, David Geffen School of Medicine, UCLA

2013 Global Health and Population Departmental Scholarship, Harvard University

2013 Deans' Health and Life Sciences Challenge Finalist, Harvard University

2013 South Asia Institute Summer Research Grant, Harvard University

2013 Maternal Health Task Force Summer Internship, Harvard University

2011 Oscar Geballe Postgraduate Scholarship, Department of Intercollegiate Athletics, UC Berkeley

2008 Outstanding Immunologist, Department of Molecular and Cell Biology, UC Berkeley

2008 Best Recruiter, Cal Men's Swim Team, UC Berkeley

Teaching and Mentoring

Fall 2020 **Tutor at Large**, *Department of Biostatistics*, Harvard University

Summer 2020 **Primary Instructor**, *Biostatistics Summer Prep Program for Incoming PhD Students*, Harvard University

2020–2021 **Student Mentor**, *Department of Biostatistics*, Harvard University

Fall 2019 **Teaching Fellow**, *Biostatistics 260: Introduction to Data Science*, Harvard University

Summer 2019 **Primary Instructor**, *StatStart*, Harvard University

Summer 2019 **Teaching Fellow**, *Data Science in Action: Convolutional Neural Networks for Self-Driving Cars*, Harvard University

Spring 2019 **Teaching Fellow**, *Biostatistics 210: Applied Regression Analysis*, Harvard University

Fall 2018 **Teaching Fellow**, *ID 201: Core Principles of Biostatistics and Epidemiology*, Harvard University

Fall 2018 **Teaching Fellow**, *Biostatistics 210: Applied Regression Analysis*, Harvard University

2017–2021 **Tutor**, *Department of Biostatistics*, Harvard University

Service and Leadership

2013–Present **Co-Founder**, *TraumaLink*, Dhaka, Bangladesh

2019–Present **Reviewer**, *Trials*

2019–2021 **Reviewer**, *Harvard Public Health Review*

2018–2021 **Consultant**, *Biostatistics Student Consulting Center*, Harvard University

2015–2016 **Co-President**, *Connecting Californians to Care*, Los Angeles, CA

2012 **Volunteer**, *Ambassadors for Sustained Health*, New York City, NY

2011–2012 **HIV Test Counselor**, *GMHC*, New York City, NY

Athletics Cal Men's Swimming Team (2003–2008); Bear Student Athlete Advisory Committee (2006–2008); Big C Society (2008–Present)

Research Experience

2013 **Research Assistant**, *Global Fund to Fight AIDS, Tuberculosis and Malaria*

2013 **Summer Fellow**, *Population Council*, Dhaka, Bangladesh

2008–2010 **Research Associate**, *Aurora Algae, Inc.*, Alameda, CA

2006–2008 **Research Assistant**, *Vance Lab, UC Berkeley*, Berkeley, CA